SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 2 – Concept Mapping
Weightage:	20% of CIA	**Total Marks:**	100
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
Student Name:	M. Rakesh	**Reg. No.:**	192421006
Section / Batch:	5T	**Date:**	1/8/26

Instructions to Students

- Answer the following question. Total Marks: 100.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Question Type	Focus Area	Questions
Concept Mapping	Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method	Q1

SECTION A – CONCEPT MAPPING

Code of Conduct

I M. Rakesh (192421006) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: M. R

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%				
Linkages	25%				
Organization	20%				
Integration of Literature	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: ____________	Signature: ____________	Signature: ____________
Date: ____________	Date: ____________	Date: ____________

Master Theorem - Case Selection Map.

$T(n) = aT(n/b) + f(n)$. $(a \geq 1,\ b > 1,\ f(n) > 0)$

Key Comparison:
Compare $f(n)$ with $n^{\log_b a}$

Case	Comparison $f(n)$ vs $n^{\log_b a}$	How Case is Selected.	Solution
Case 1 f(n) grows polynomially slower	$f(n) = O(n^{\log_b a - \varepsilon})$ for some $\varepsilon > 0$)	* Divide $f(n)$ by $n^{\log_b a}$ * If limit $= 0$ $f(n)$ is asymptotically smaller	$T(n) = \Theta(n^{\log_b a})$
Case 2 f(n) grows at same rate	$f(n) = \Theta(n^{\log_b a})$	* Divide $f(n)$ by $n^{\log_b a}$ if limit $= c$ $(0 < c < \infty)$ same order.	$T(n) = \Theta(n^{\log_b a} \log n)$
Case 3 f(n) grows polynomially faster	$f(n) = \Omega(n^{\log_b a + \varepsilon})$ for some $\varepsilon > 0$ and $a f(n/b) \leq c f(n)$ for some $c < 1$	* Divide $f(n)$ by $n^{\log_b a}$ * If limit $= \infty$ $f(n)$ is asymptotically larger	$T(n) = \Theta(f(n))$

$f(n) = O(n^{\log_b a - \varepsilon})$ $\longleftarrow$ $f(n) = \Theta(n^{\log_b a})$ $\longrightarrow$ $f(n) = \Omega(n^{\log_b a + \varepsilon})$

Smaller. (same). (larger)

$\downarrow$

$n^{\log_b a}$